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Welcome to the Lab n°3 :)



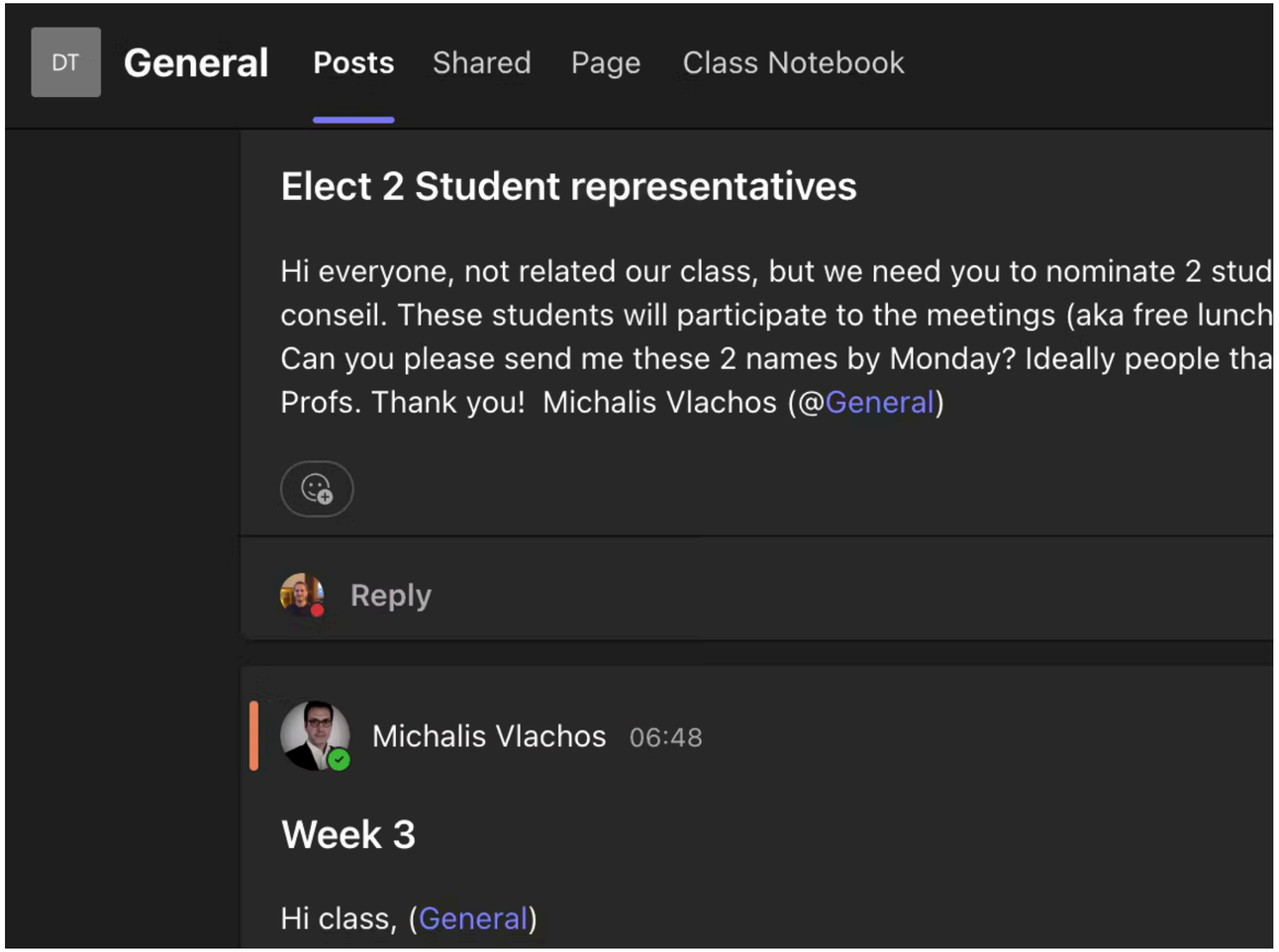
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Today's Lab Structure

- → Walkthrough : classification
- → Pause @17h00
- → Assignement 3
- → Feedback on Assignment 1

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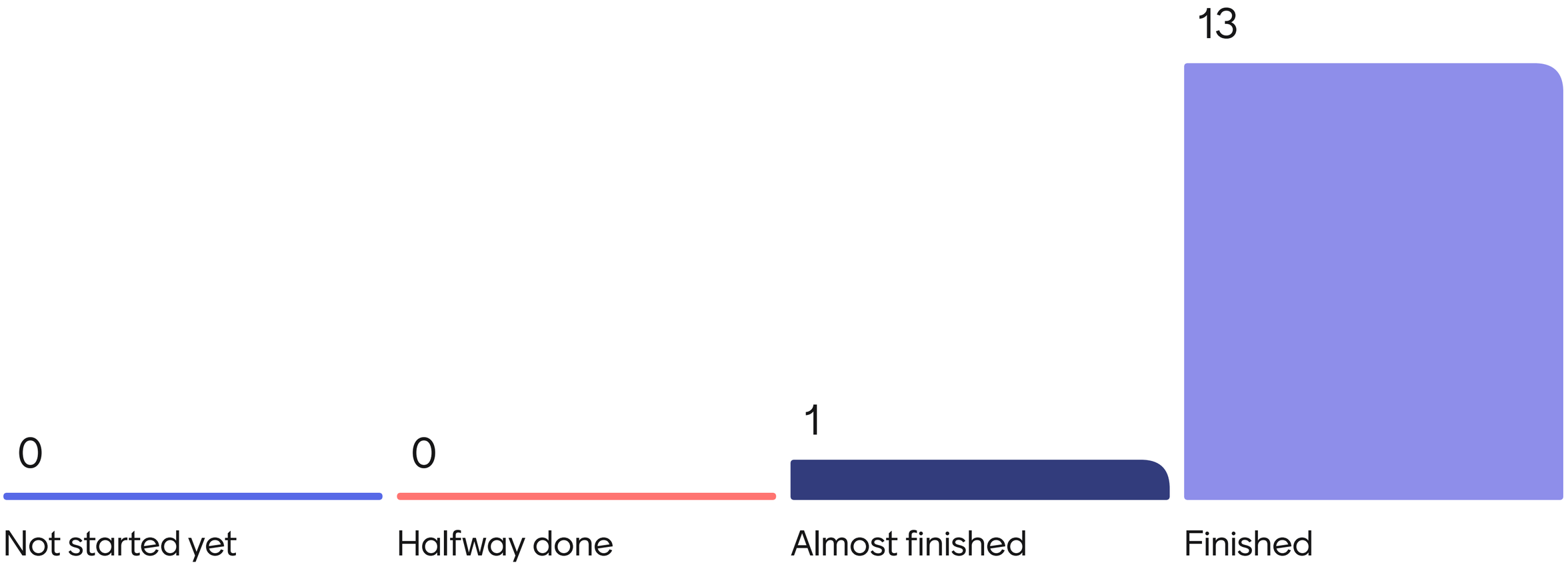


Does everyone have access to the Teams channel?

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Where are you in the progress of Assignment 2?



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How would you rate your experience with the Assignment 2?

0	I need help: It was hard to follow and I need more guidance
0	Challenging: I struggled with multiple sections
0	Okay: Some sections were confusing
1	Good: Mostly clear, but a few tricky parts
0	Excellent: Everything was clear and easy to follow



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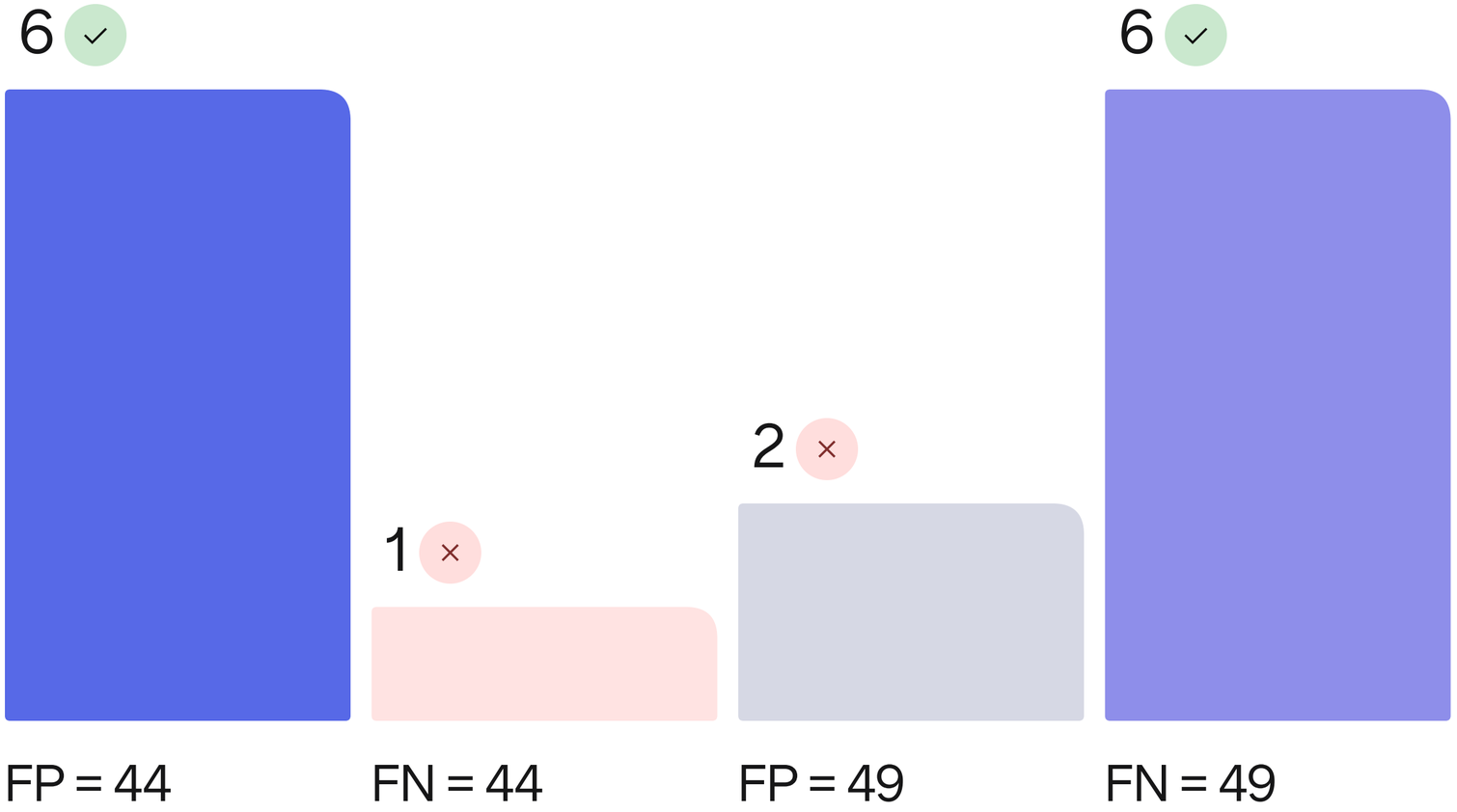
Walkthrough : Classification

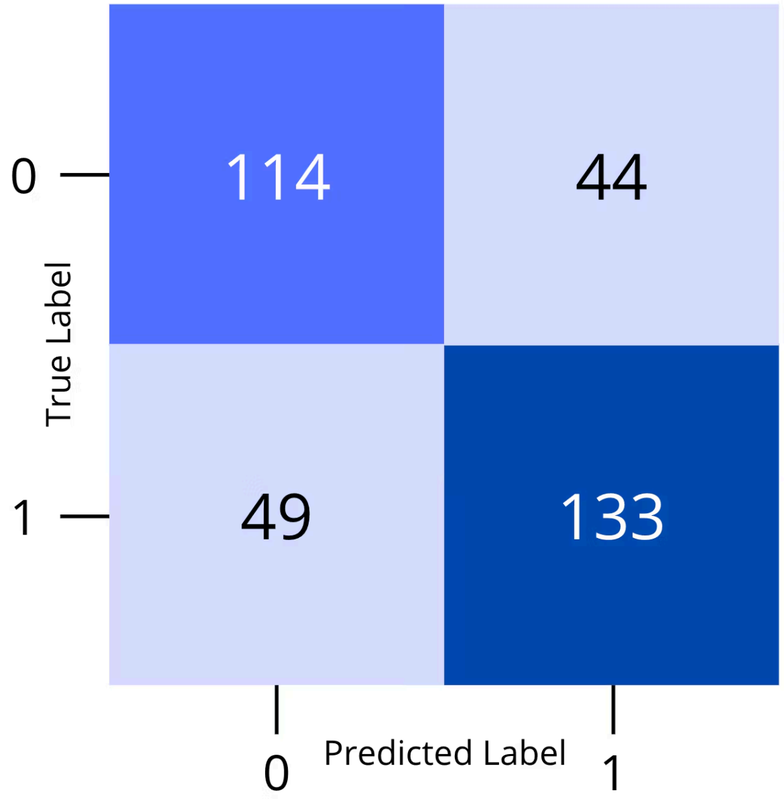
- Logistic Regression
- Evaluating classifier
- Cross-Validation
- Your Turn !
- Decision Tree
- KNN

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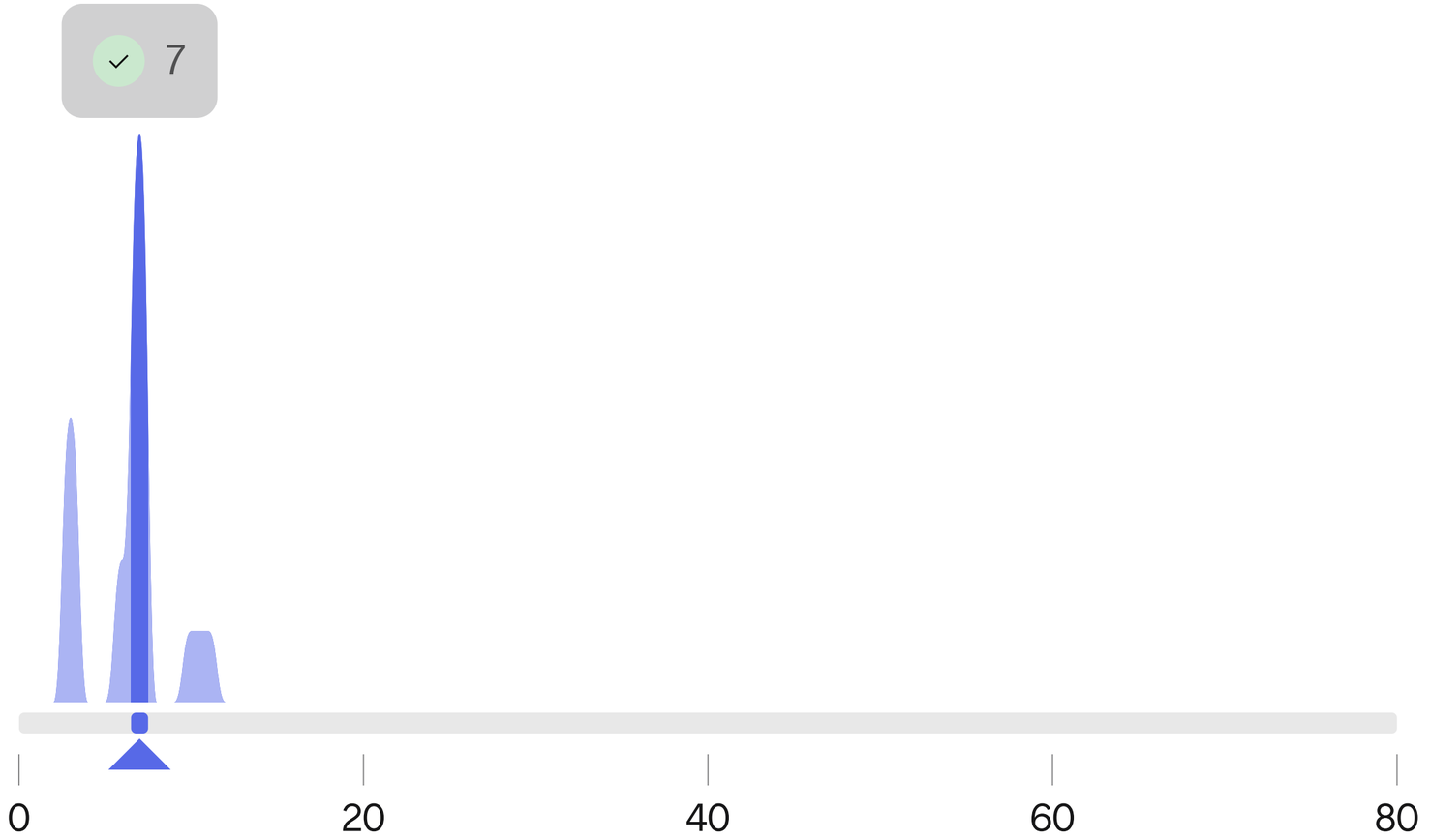
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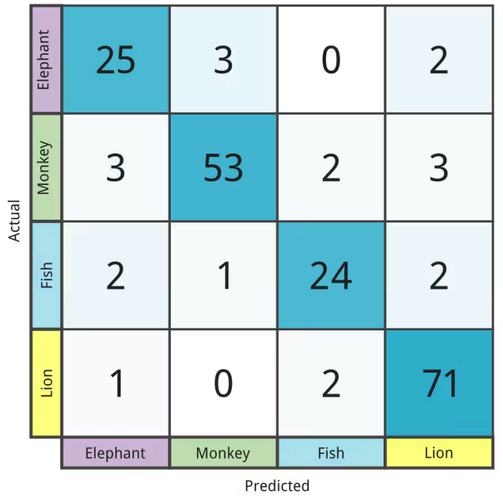
Find the False Positive (FP) and False Negative (FN).



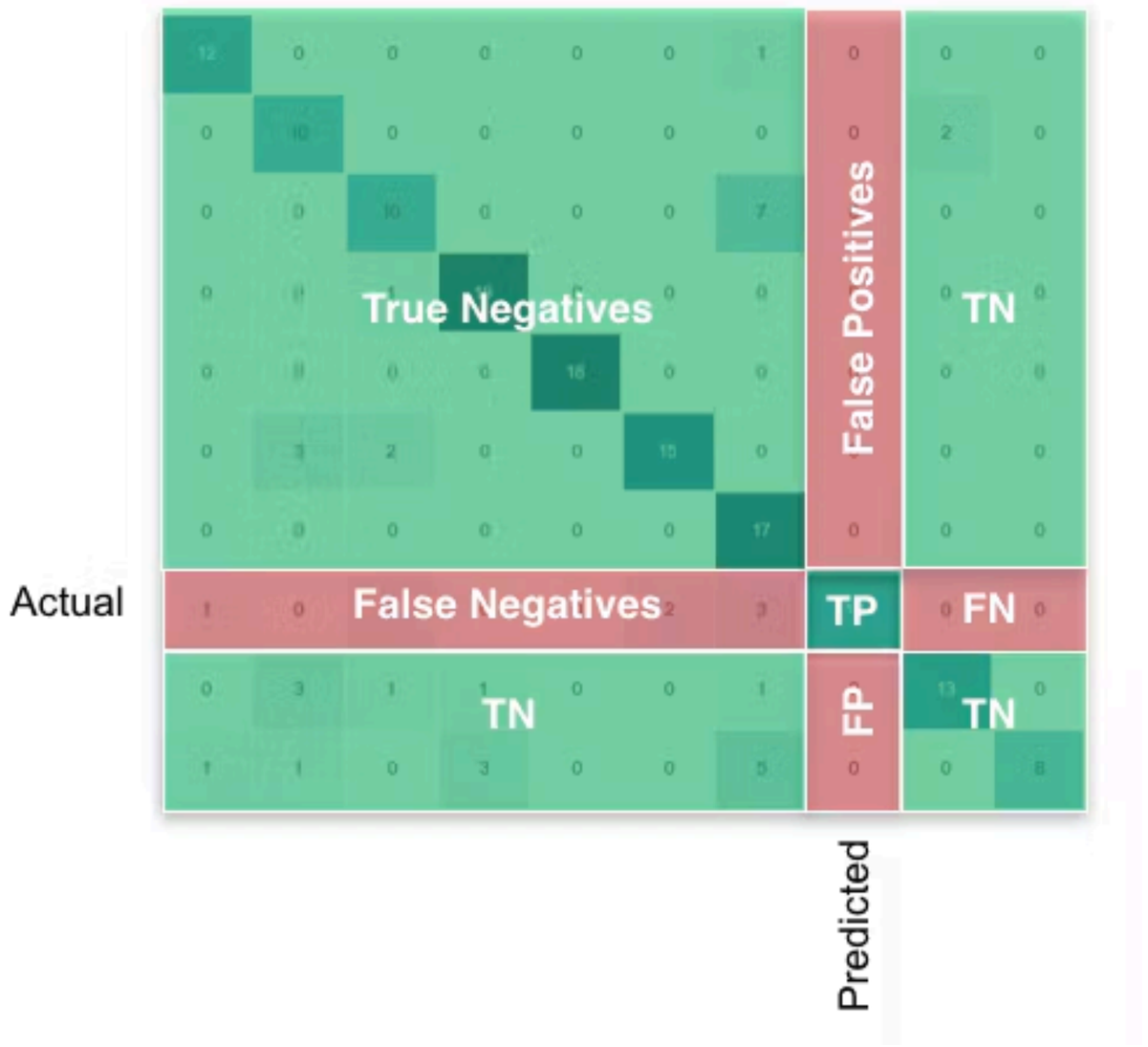


How many FP are there for lion ? (In yellow)

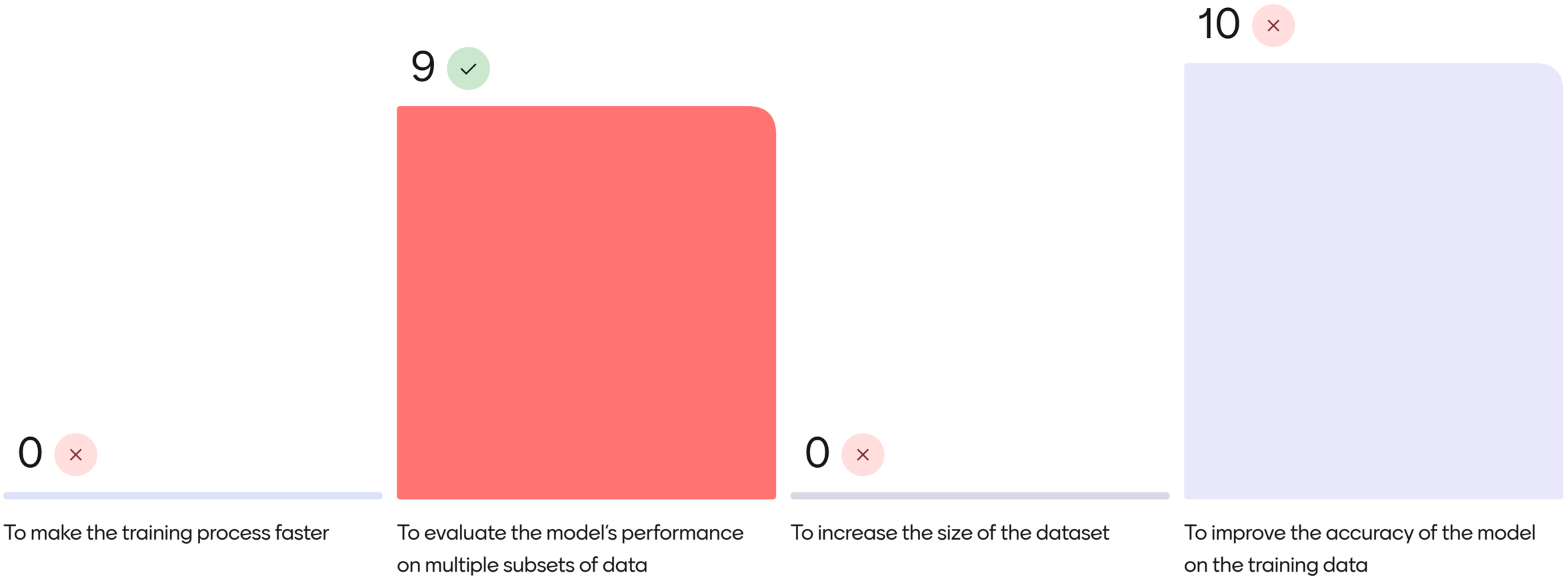




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True Negatives
False Positives
TN
Actual
False Negatives
TP
FN
TN
FP
TN
Predicted

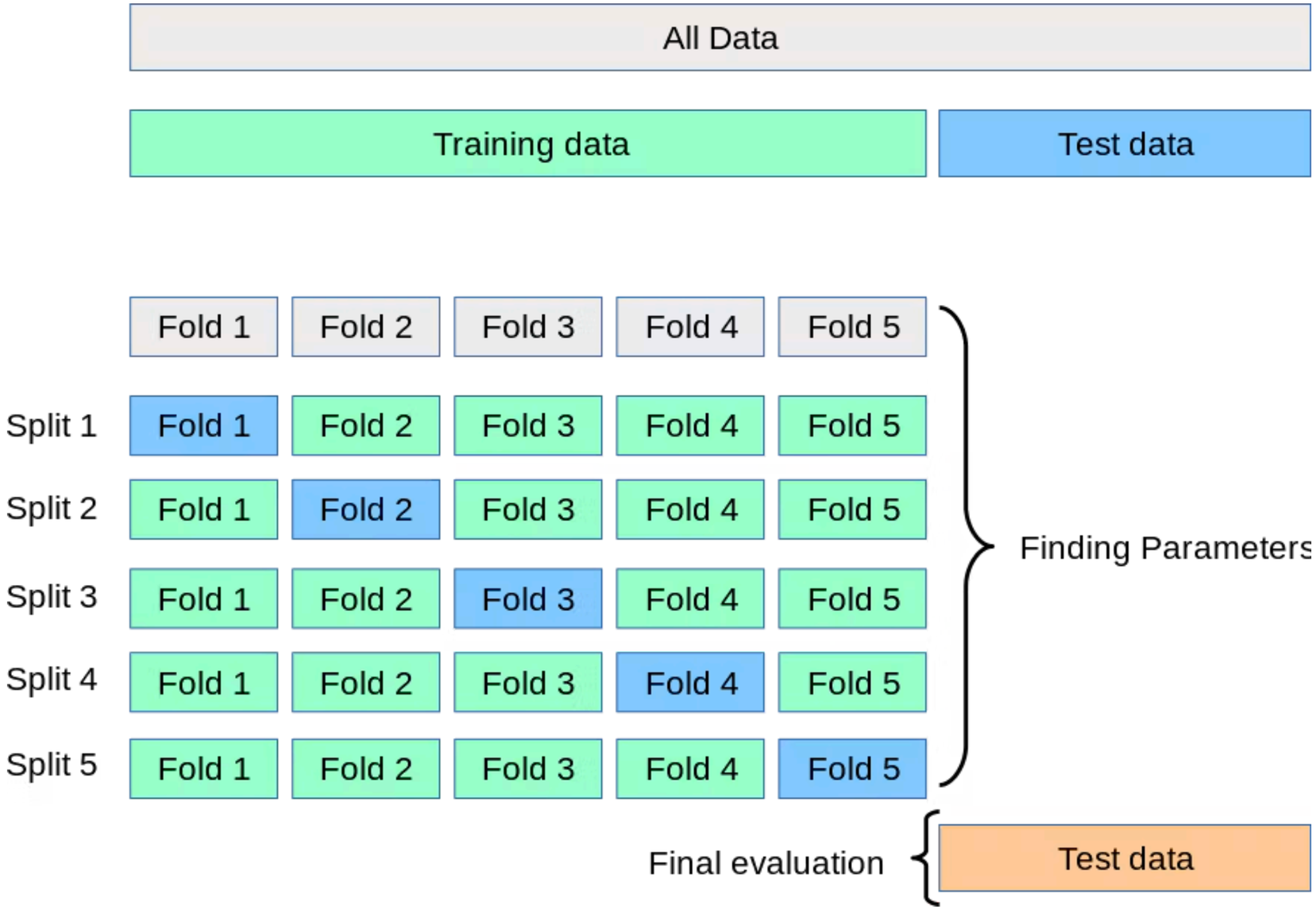
What is the main purpose of using cross-validation in machine learning?



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Imagine you're practicing for an exam, and you have **5 different practice tests**. Instead of studying for only one test and checking how well you did, you **practice on 4 of them** and use the **5th one to check your score**. You do this **5 times**, each time using a different practice test as your "test" and the other 4 for "study". At the end, you **average your scores from all 5 tests** to get a better idea of how well prepared you are.

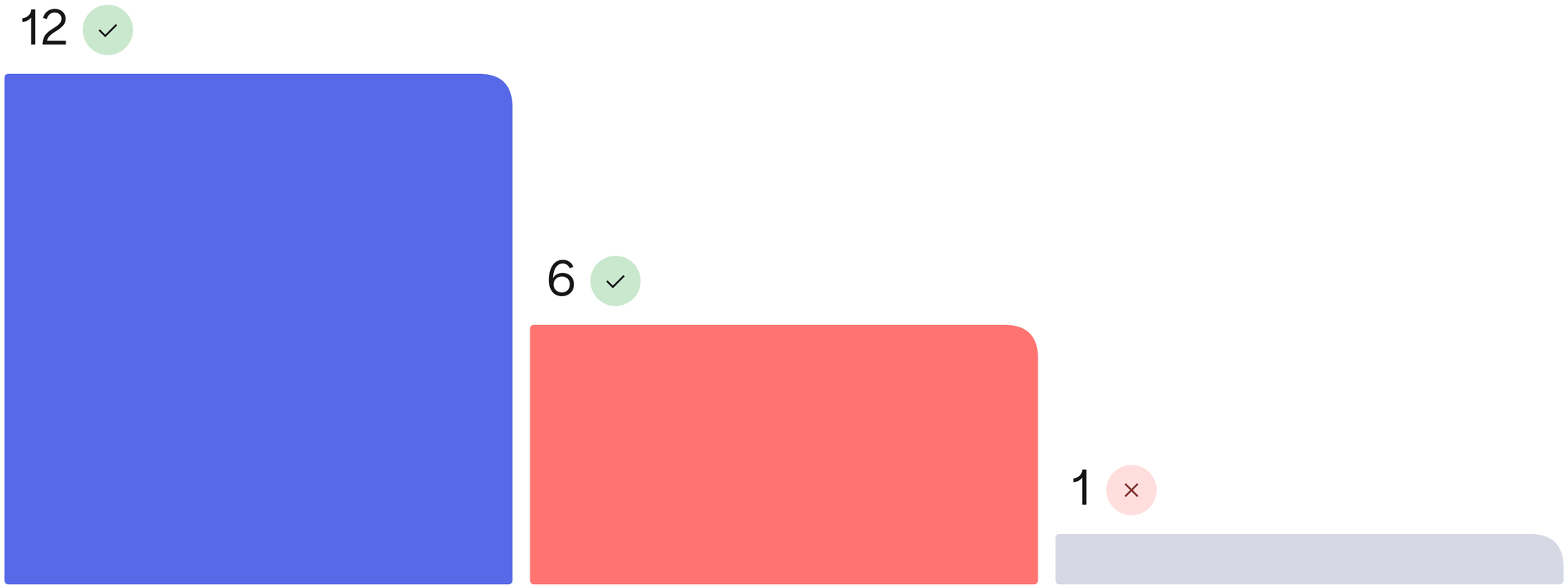
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All Data
Training data
Test data
Fold 1
Fold 2
Fold 3
Fold 4
Fold 5
Split 1
Split 2
Split 3
Split 4
Split 5
Finding Parameters
Final evaluation
Test data

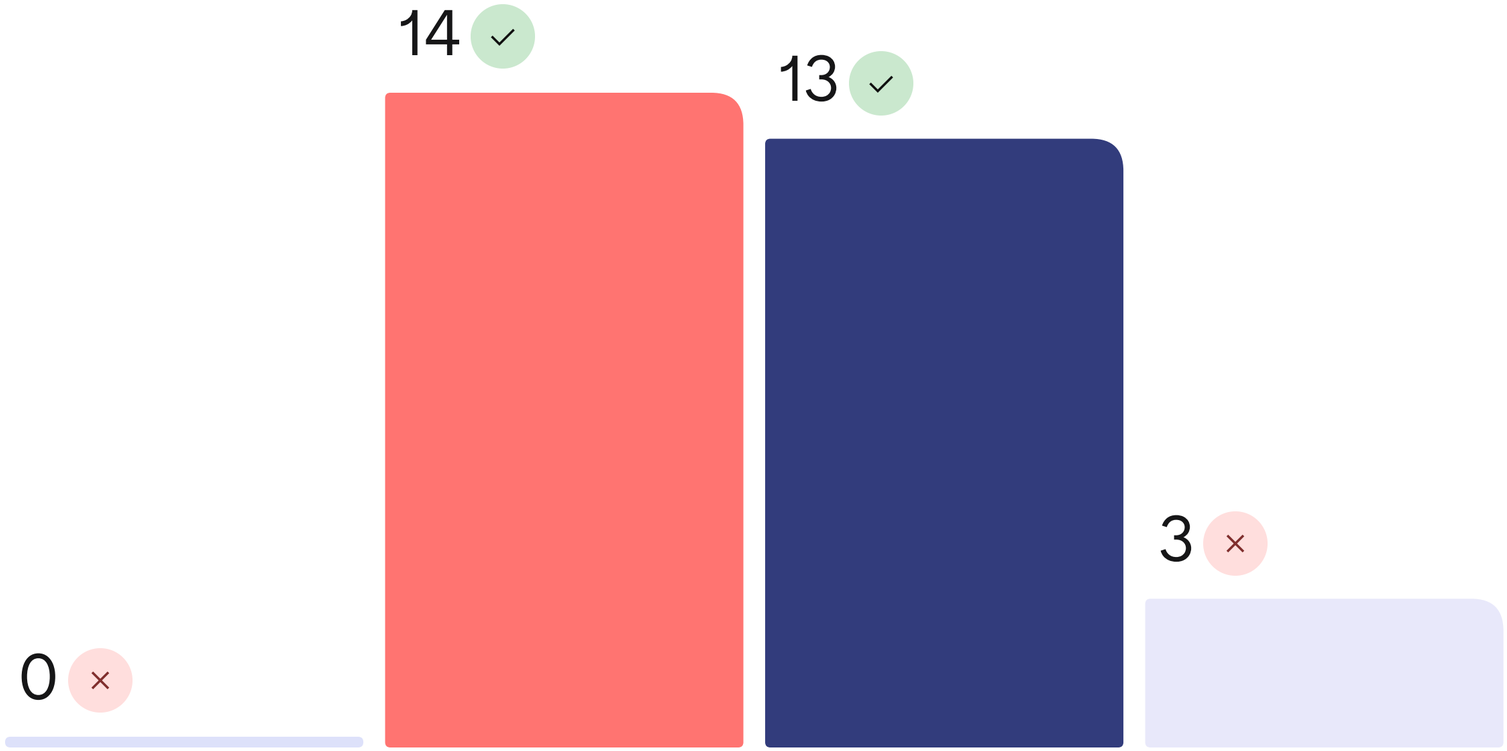
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How does the **max_depth** parameter affect the performance of a decision tree?



kNN - What is true ? (level 1)

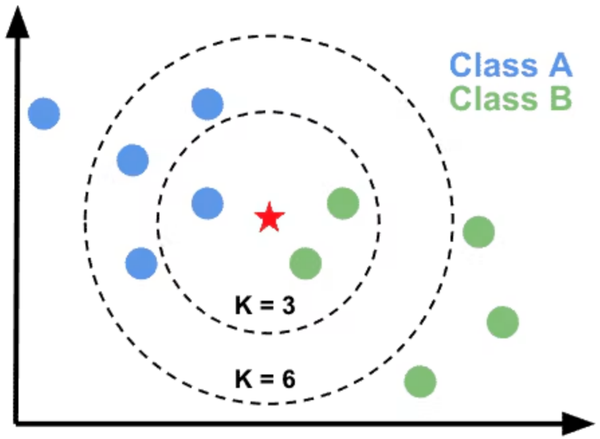


K = 3 : Star is predicted class A

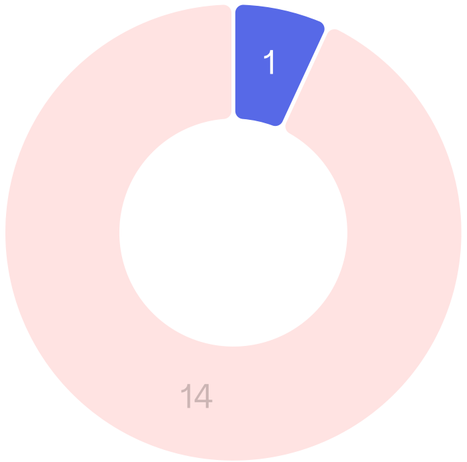
K = 3 : Star is predicted class B

K = 6 : Star is predicted Class A

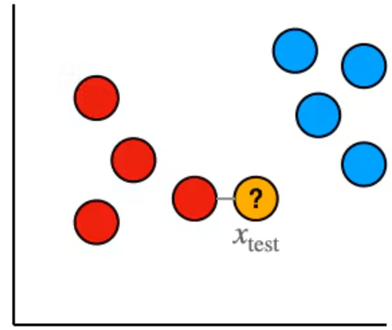
K = 6 : Star is predicted class B



kNN - What is true ? (Level 2)

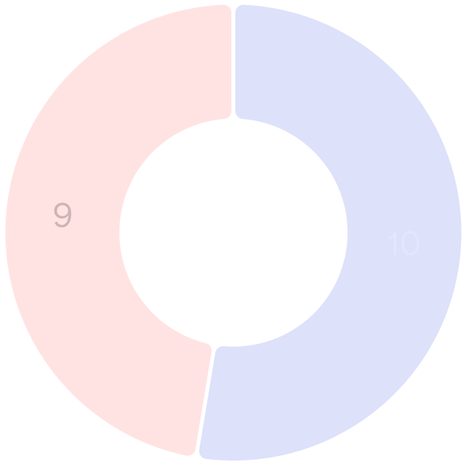


1	K = 3 : Predicted blue	✓
14	K = 3 : Predicted red	×

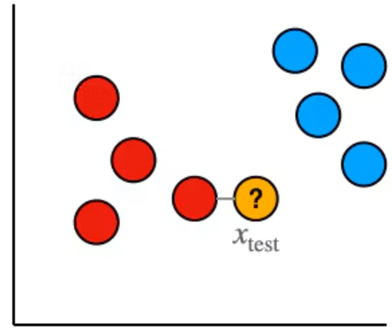


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kNN - What is true ? (Level 3)

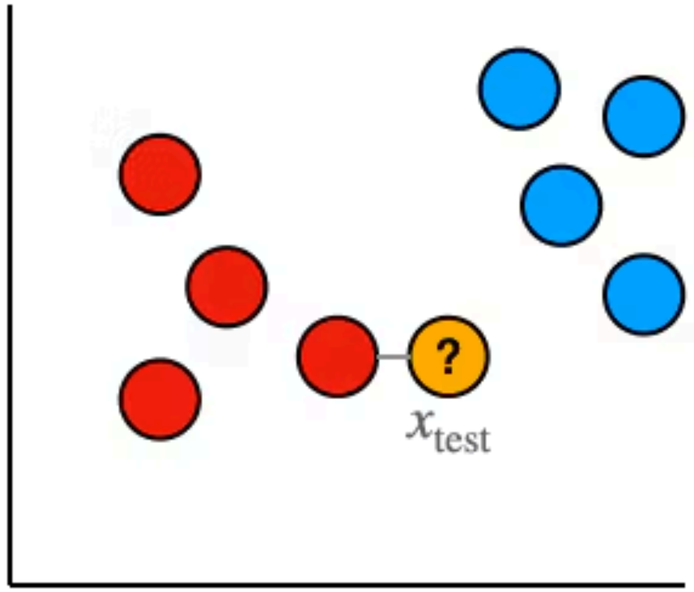


10	K = 8 : Predicted blue	×
9	K = 8 : Predicted red	×



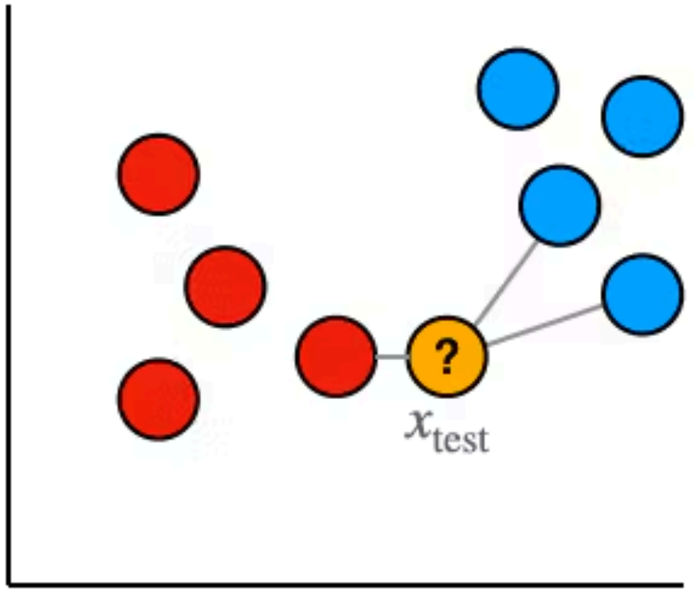
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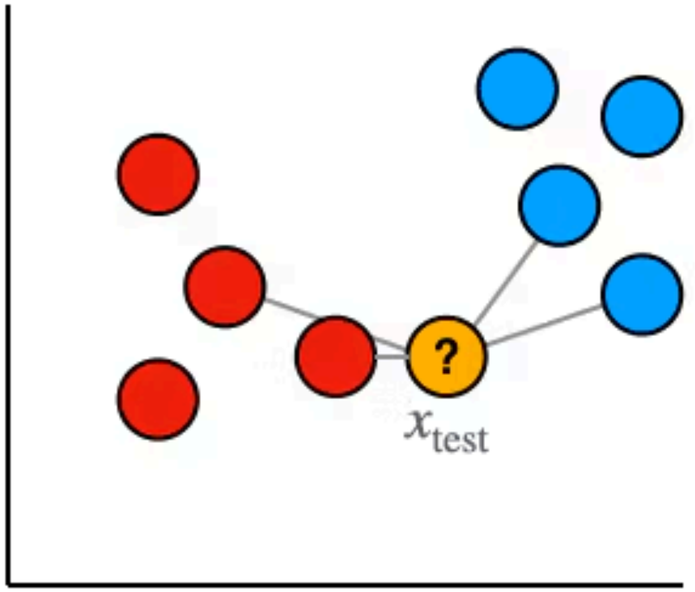
$k = 1$

Nearest point is red, so x_{test} classified as red



$k = 3$

Nearest points are {red, blue, blue} so x_{test} classified as blue



$k = 4$

Nearest points are {red, red, blue, blue} so classification of x_{test} is not properly defined

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Important informations

- Don't forget to **submit your code** (Code Q2)
- Thank you :)

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Any questions for us? Any requests? Feedback?

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